

Challenge (Habanero)

Q1. Solve for x in $2x + 5 = 11$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. A recipe calls for 3 cups of flour to make 12 cookies. If you want to make 24 cookies, how many cups of flour will you need, x ?

- (A) $x = 4$
- (B) $x = 5$
- (C) $x = 6$
- (D) $x = 7$
- (E) $x = 8$

Q3. Solve for y in $y - 2 = 7$

- (A) $y = 5$
- (B) $y = 6$
- (C) $y = 7$
- (D) $y = 8$
- (E) $y = 9$

Q4. A construction worker needs to mix 2 bags of concrete with 5 gallons of water. If he has 15 gallons of water, how many bags of concrete can he mix, x ?

- (A) $x = 6$
- (B) $x = 7$
- (C) $x = 8$
- (D) $x = 9$
- (E) $x = 10$

Q5. Solve for z in $z/4 = 9$

- (A) $z = 32$
- (B) $z = 36$
- (C) $z = 40$
- (D) $z = 44$
- (E) $z = 48$

Q6. A chef needs to package 48 cupcakes into boxes that hold 6 cupcakes each. How many boxes will she need, x ?

- (A) $x = 6$
- (B) $x = 7$
- (C) $x = 8$
- (D) $x = 9$
- (E) $x = 10$

Q7. Solve for w in $w + 2 = 11$

- (A) $w = 7$
- (B) $w = 8$
- (C) $w = 9$
- (D) $w = 10$
- (E) $w = 11$

Q8. A builder is designing a house with a rectangular garden. The length of the garden is 5 meters more than the width, x . If the perimeter of the garden is 30 meters, what is the width?

- (A) $x = 5$
- (B) $x = 6$
- (C) $x = 7$
- (D) $x = 8$
- (E) $x = 9$

Open Ended Questions (FRQ)

Q9. Solve for x in $2x + 5 = 3x - 2$ and check your solution for accuracy.

Q10. A chef is making a recipe that calls for $2/3$ cup of sugar. If she wants to make 3 times the recipe, how much sugar will she need, x ? Show your work and check your solution for accuracy.

Chef's Tips

- Check for accurate calculations
- Verify the units of measurement
- Ensure the solutions are reasonable in the context of the problem